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### HAIRBRUSH APPLICATOR AND HAIRBRUSH DISPENSER CONFIGURATIONS

#### Abstract

Configurations are provided for hairbrush applicators and personal care devices, including dispensing and application of material to a user, such as powders and dry shampoo. According to embodiments, a hairbrush applicator includes a bristle assembly having bristles and a base tray, the base tray including at least one channel opening. The hairbrush applicator includes an applicator base configured to receive the bristle assembly and at least one flow hole to provide material to the bristles. The applicator base includes a rim with a guide channel, a collar is configured to engage with the rim and the guide channel. The collar can rotate relative to the applicator base to provide at least one position to dispense material to the bristle assembly. The applicator base may be configured to engage with a storage container configured to store material dispensed by the applicator which may include a disposable cartridge.

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## Background/Summary

### FIELD

[0001] The present disclosure generally relates to personal care products including hairbrushes and cosmetic applicators, and more particularly to applicator elements and configurations.

### BACKGROUND

[0002] Personal care products are produced in many configurations for various forms of self-care and a wide array of uses. Hairbrushes, for example, come in many designs and configurations for styling and brushing hair. Hairbrushes and combs often include a handle and arrangement of tines to brush hair. Despite various existing hairbrush configurations and cosmetic products in general, there exists a need for haircare products that assist with application of products to hair. By way of example, dry shampoo applicators can include pads for receiving product. These products often do not allow for proper application and can result in too much product being applied to a user. Other products include aerosol sprays. Due to byproducts of the spray, there exists a desire to reduce the use of aerosol products including spray on dry shampoos and spray on products. Aerosol sprays also may output an excess amount of material resulting in too much product being applied to a user. Application of an aerosol spray may also result in unwanted material being sprayed on a user's face, clothes, and surrounding areas. There exists a need which reduces the use of aerosol sprays and minimizes waste of personal care products.

[0003] There exists a need for applying product in a targeted way so as to apply powder in desired amounts and areas. There exists a need for applying product in a way that is easily distributed without the need for using a brush or comb to get the powder dispersed through the hair after powder application (or simultaneously). There exists a need for separating the hair strands while applying powder so that the powder can be dispersed evenly. There exists a need for limiting the powder from coming into contact with the scalp so that the powder does not cause scalp itchiness, clogged pores, or have to be washed out after each use. There exists a need for compact, travel-friendly, applicator that doesn't require the use of flammable liquids for passing TSA regulations. There also exists a need for products and designs to allow for application of dry shampoo and personal care products.

### BRIEF SUMMARY OF THE EMBODIMENTS

[0004] Disclosed and described herein are configurations for hairbrush applicators and personal care applicators. According to embodiments, a hairbrush applicator configuration is provided including a bristle assembly including a plurality of bristles and a base tray, wherein the plurality of bristles are secured to the base tray, and wherein the base tray includes at least one channel opening. The hairbrush applicator configuration includes an applicator base configured to receive the bristle assembly, the applicator base including a body element having at least one flow hole, wherein the at least one flow hole is configured to provide material to the bristle assembly, wherein the applicator base includes a rim including at least one guide channel. The hairbrush applicator configuration includes a collar configured to engage with the rim, wherein the collar is configured to engage with the at least one guide channel, and wherein the collar is configured to rotate relative to the applicator base to provide at least a first position and a second position, wherein the second position is configured to arrange the at least one channel opening of the base tray and the at least one flow hole of the body element to dispense material to the bristle assembly.

[0005] According to embodiments, the plurality of bristles includes a first set of bristles including an arrangement of fiber elements, and a second set of bristles including a plurality of bristles each having a shaft and molded tip.

[0006] According to embodiments, the plurality of bristles are grafted to the base tray.

[0007] According to embodiments, the collar includes at least one tab protruding from an inner surface of the collar, the at least one tab configured to engage with the at least one guide channel.

[0008] According to embodiments, the base tray of the bristle assembly includes at least one notch and the collar includes at least one protrusion configured to engage with the at least one notch of the base tray, and wherein the collar is configured to rotate the bristle assembly to align of the collar the at least one channel opening with the at least one flow hole.

[0009] According to embodiments, the collar is configured to rotate within the same plane of the at least one guide channel, wherein the first position is configured as an off position, an intermediate position of the collar is a first flow setting, and the second position of the collar is a second flow setting, wherein the second flow setting provides increased flow of material relative to the first position.

[0010] According to embodiments, the body element of the applicator base includes a plurality of flow holes having a first opening size, and a plurality of flow holes having a second opening size, wherein the second opening size is greater than the first opening size.

[0011] According to embodiments, the applicator base includes a notch opening for each of the first position and the second position, and a slot between the first and the second position.

[0012] According to embodiments, the applicator base is configured to engage with a storage container, the storage container including a compartment configured to store material dispensed by the applicator.

[0013] According to embodiments, the applicator base and collar form a handle for the hairbrush applicator, and the bristle assembly is configured to apply at least one of dry shampoo and powder to user hair.

[0014] Embodiments are also directed to a hairbrush applicator including a storage container. The hairbrush applicator includes a bristle assembly including a plurality of bristles and a base tray, wherein the plurality of bristles are secured to the base tray, and wherein the base tray includes at least one channel opening. The hairbrush applicator includes a storage container including a compartment configured to store material dispensed by the hairbrush applicator. The hairbrush applicator includes an applicator base configured to receive the bristle assembly, the applicator base including a body element having at least one flow hole, wherein the at least one flow hole is configured to provide material to the bristle assembly, wherein the applicator base includes a rim including at least one guide channel. The hairbrush applicator includes a collar configured to engage with the rim, wherein the collar is configured to engage with the at least one guide channel, and wherein the collar is configured to rotate relative to the applicator base to provide at least a first position and a second position, wherein the second position is configured to arrange the at least one channel opening of the base tray and the at least one flow hole of the body element to dispense material to the bristle assembly.

[0015] According to embodiments, the applicator base, storage container and collar form a handle for the hairbrush applicator, and the bristle assembly is configured to apply at least one of dry shampoo and powder to user hair.

[0016] According to embodiments, the storage container is configured to receive and retain a removable cartridge including the material to dispense to the bristle assembly.

[0017] Other aspects, features, and techniques will be apparent to one skilled in the relevant art in view of the following detailed description of the embodiments.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The features, objects, and advantages of the present disclosure will become more apparent

from the detailed description set forth below when taken in conjunction with the drawings in which like reference characters identify correspondingly throughout and wherein:

[0019] FIGS. 1A-1C illustrate a hairbrush applicator according to one or more embodiments;

[0020] FIG. 2A illustrates a hairbrush applicator with a container according to one or more embodiments;

[0021] FIG. 2B illustrates an exploded view of a hairbrush applicator with a container according to one or more embodiments;

[0022] FIGS. 3A-3B illustrate a hairbrush applicator collar according to one or more embodiments;

[0023] FIGS. 4A-4B illustrate a hairbrush applicator bristle assembly according to one or more embodiments;

[0024] FIGS. 5A-5C illustrate an applicator base of a hairbrush applicator according to one or more embodiments; and

[0025] FIGS. 6A-6B illustrate a material cartridge according to one or more embodiments.

## DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

### Overview and Terminology

[0026] Embodiments of the disclosure are directed to configurations for personal care products including hairbrush applicators and dry shampoo brushes. According to embodiments, configurations provide features for application of a product or material to a user, such as a powder, dry shampoo or personal care product in general. Embodiments provide configurations that allow for a plurality of flow control settings and a brush configuration to evenly distribute material to a user's hair. By controlling flow of material, users can evenly distribute products and limit and/or avoid waste of a product. In addition, hairbrush applicators may use products that do not require aerosol sprays for application of a product.

[0027] Haircare products, such as dry shampoo, may be used to boost texture and volume of hair. Dry or powder products may be difficult to apply for some users. Embodiments provide configurations that include hairbrush applicators that are fast acting and easy to use. Configurations can limit and/or eliminate residue. In addition to application of dry shampoo material to a user hair, such as a user scalp, hairbrush applicators as described herein may include one or more features to clean and remove oil from a user's hair. By way of example, hairbrush applicators may be used to apply product to a user's hair, distribute the product by using the hairbrush application to brush hair and removal oil from a user's hair or scalp. Hairbrush applicators according to embodiments may also be used for leave in products.

[0028] According to embodiments, a hairbrush applicator may include a bristle assembly and one or more structural elements. Structural elements may allow for the hairbrush applicator to be attached to one or more styles of container, such as a powder or dry shampoo container. In addition, structural elements may allow for providing a configuration that allows for one or more positions for controlling flow of material to bristles, such as first setting as an “off” position, a second setting to allow for an “on” position and one or more intermediary positions. One or more positions may provide a level of flow control. Structural elements of hairbrush applicators according to embodiments, allow for a compact design, a recyclable product and configurations that allow for dispensing and retaining material.

[0029] According to embodiments, a hairbrush applicator may include bristles and structural elements that control the flow of material to bristles. According to embodiments, hairbrush applicator configurations may include a container coupled to one or more structural elements of the hairbrush applicator. Embodiments describe configurations for containers and use of disposable or refillable cartridges. It should be appreciated that hairbrush applicators may be used with one or more container configurations.

[0030] According to embodiments, a hairbrush applicator may include a bristle assembly, an applicator base, and a collar. The applicator base may be configured to receive and/or support the bristle assembly, and the collar may be configured to be coupled to the applicator base. The collar

and the applicator base may retain the bristle assembly, and rotation of the collar may result in one or more flow holes of the applicator base and channels of the bristle assembly to be aligned. By aligning openings in the applicator base and bristle assembly, material may flow from the applicator base to bristles of the bristle assembly. According to embodiments, flow control may be provided by a plurality of openings, such as flow holes in an applicator base. An applicator base may include one or more sets or groups of flow openings. Configurations may include one or more structural elements to align flow holes.

[0031] According to embodiments, hairbrush applicators may be configured to use one or more refills, capsules, pouches and/or cartridges. Embodiments provide solutions for user action to pour or refill loose material in a container. Alternatively, or in addition, embodiments provide solutions that reduce the amount of material needed and material that may be wasted after use. By way of example, a hairbrush applicator configuration may be configured to receive a pouch or cartridge or material that is sealed, and which may use less packaging than a replacement container.

Replaceable cartridges or pouches may be an inner cartridge or container that includes a leak protection seal that allows for transport/shipping and which may be removed in full, or in part, prior to use with a hairbrush applicator.

[0032] Embodiments provide solutions for application of products including dry shampoo and powder materials. According to embodiments, configurations include one or more elements to apply product in a targeted way, such as applying powder in desired amounts and areas, and for applying product that is easily distributed without the need for using a separate brush or comb to get the powder dispersed through the hair after powder application. Embodiments provide elements configured to separate hair strands while applying powder so that the powder can be dispersed evenly. In addition, embodiments provide configurations for limiting powder from coming into contact with the scalp so that the powder does not cause scalp itchiness, clogged pores, or have to be washed out after each use. Embodiments also provide compact, travel-friendly, applicators that do not require the use of flammable liquids for passing TSA regulations.

[0033] It should be appreciated that embodiments are not limited to personal care products or hairbrush applicators and may be used and/or applied in other spaces. By way of example, embodiments may provide configurations for one or more of a scalp applicator, scalp massager, dry shampoo brush, haircare product dispenser, hairbrush bristles, brushes in general and other personal care devices. Similarly, embodiments may provide configurations for application of one or more of powders, personal care products, minerals, mineral mixtures and other types of material.

[0034] As used herein, the terms “a” or “an” shall mean one or more than one. The term “plurality” shall mean two or more than two. The term “another” is defined as a second or more. The terms “including” and/or “having” are open ended (e.g., comprising). The term “or” as used herein is to be interpreted as inclusive or meaning any one or any combination. Therefore, “A, B or C” means “any of the following: A; B; C; A and B; A and C; B and C; A, B and C”. An exception to this definition will occur only when a combination of elements, functions, steps or acts are in some way inherently mutually exclusive.

[0035] Reference throughout this document to “one embodiment,” “certain embodiments,” “an embodiment,” or similar term means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of such phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner on one or more embodiments without limitation.

#### Exemplary Embodiments

[0036] FIGS. 1A-1C illustrate hairbrush applicator **100** according to one or more embodiments. FIG. 1A shows a side-profile view of hairbrush applicator **100** including bristle assembly **105**, applicator base **110** and collar **115**. According to embodiments, hairbrush applicator **100** may be configured to apply material, such as a powder, dry shampoo, personal care product in general, to a

user. Hairbrush applicator **100** includes structural features which allow for control and dispensing material which are not provided by conventional devices. Similarly hairbrush applicator **100** may include one or more positions or settings that allow for different operation relative to existing devices. In addition, hairbrush applicator **100** includes a design and configuration that reduces the amount of plastic used to form the device. Moreover, hairbrush applicator **100** includes a design and configuration that allows for improved manufacturing by decreasing the need for injection molding.

[0037] According to embodiments, hairbrush applicator **100** is an applicator for one or more personal care products, the products may be applied by way of bristle assembly **105**. Bristle assembly **105** may include at least one set of bristles, such as bristles **106**. One or more components of hairbrush applicator **105** may allow for material to be dispensed or released to bristle assembly **105** and bristle assembly **105** may be used as a brush to apply product dispensed to bristles to a user. By way of example, for a powder or dry shampoo, product may be released to bristles **106** of bristle assembly **105** and bristles **106** may be used to brush a user's hair and/or scalp. Portions of hairbrush applicator **105** may be used as a handle, such as one or more of collar **115** and applicator base **110**. Portions of hairbrush applicator **105**, such as one or more of collar **115** and applicator base **110**, may be used to adjust the rate material is dispensed or released to bristles **106**. According to embodiments, collar **115** may rotate relative to applicator base **110** from a first position (e.g., off position) to at least one of an intermediate position (e.g., light flow) and a second position (e.g., regular, heavy or increased flow). According to embodiments, rotation of collar **115** to one or more positions may be settings for dispensing material. In addition, each rotation position may relate to one or more of a size and amount or flow holes or channels for material to flow or pass through to bristles **106**. Hairbrush applicator **100** may include one or more elements to allow for collar **115** to be rotated relative to applicator base **110** and to allow collar **115** to be coupled to applicator base **110**.

[0038] According to embodiments, hairbrush applicator **100** is a cap or topper for at least one of a container or storage element. Applicator base **110** may be configured to secure hairbrush applicator to the container. Coupling of an applicator base to a container is shown and described herein below with reference to FIGS. 2A-2B. According to embodiments, when hair brush applicator **100** is in an open position, material such as dry shampoo or other personal care products may be dispensed from the container to bristle assembly **105** by inverting or turning over the hairbrush applicator. Material may be dispensed while using the hairbrush applicator. For example, a user may hold the hairbrush applicator in a position, such as an inverted, angled or tilted position to use bristle assembly **105** to brush the user's hair. Material dispensed to the bristle assembly and/or in the container may be applied to the user's hair and/or scalp. Although embodiments of hairbrush applicator **100** are described as applying dry shampoo, it should be appreciated that other types of material may be dispensed and for other personal care purposes and applications in general.

[0039] According to embodiments, bristle assembly **105** includes a plurality of bristles, including bristles **106**, and at least one structural element, such as a base tray securing the bristles. Bristles may be mounted in the structural element such that bristles are secured and allow for material to pass through one or more channel openings in the base tray to bristles **106**. Bristles **106** may be fibers similar to hairs. Bristles **106** may be densely packed and stand upright. Bristle assembly **105** may include rigid and/or soft bristles used in hair care for smoothing, styling, and detangling human hair, or for grooming. Bristles assembly **105** may limit material, such as powder, from coming into contact with a user's scalp so that the powder does not settle into the pores causing adverse reactions like scalp itchiness, clogged pores, or temporary hair loss. Bristles assembly **105** may include a second set of bristles including a shaft portion and molded end. According to embodiments, bristle assembly **105** may be received and/or supported by applicator base **110**. A structural element, such as a base tray, may be received by a support surface of applicator base **110**. According to embodiments, collar **115** may interface with applicator base **110** to secure bristle

assembly **105**. As such, collar **115** and applicator base **110** may retain bristle assembly **105**. FIGS. **4A-4B** illustrate features of bristles assemble **105** according to one or more embodiments.

[0040] FIG. **1B** illustrates a plan view of hairbrush applicator **100** including bristle assembly **105** and collar **115**. According to embodiments, bristle assembly **105** includes a first set of bristles, bristles **106** including an arrangement of fiber elements. Bristle assembly **105** includes a second set of bristles **1071-n**. Bristles **1071-n** may each include a plurality of bristles each having a shaft and molded tip. According to embodiments, bristles **1071-n** include a pattern or arrangement of bristles in a two, four, four, two pattern that includes a row of two bristles, two rows of four bristles and a row of two bristles as shown in FIG. **1B** with respect to the bottom to top of the bristle assembly **105**. Bristles **106** may be soft or fiber type bristles to gently apply material to user hair, distribute material to user hair and remove excess material from user hair. Bristles **1071-n** may be formed of plastic or other harder material relative to bristles **106**, and bristles **1071-n** may be used to brush user hair and/or massage scalp. Bristles **106** and bristles **1071-n** may each be grafted and/or inserted to a base tray of bristle assembly **105**.

[0041] FIG. **1C** illustrates a bottom view (e.g., under side) of hairbrush applicator **100** including applicator base **110**. According to embodiments, applicator base **110** is configured to receive a container and/or material. Applicator base **110** may include at least one of threads **111** and a seal **112** (e.g., gasket) to couple to a container. Threads may allow for coupling by way of rotation and/or snap connection to a container. Seal **112** may be a rubber gasket or seal in general to allow for at least one of an engagement and to prevent leakage of material. Applicator base **115** may include body element **120** having at least one flow hole configured to provide material to a bristle assembly. Applicator base **110** may include a plurality of flow holes and the flow holes may be in a pattern. According to embodiments, body element **120** includes a first set of flow holes **1251-n** and a second set of flow holes **1301-n**. First set of flow holes **1251-n** may be a first size and second set of flow holes **1301-n** may be a second size, such as a large size to allow for increased flow and/or release of material. According to embodiments, flow holes **1251-n** and **1301-n** may be arranged in a circular pattern such that rotation of a collar (e.g., collar **110**) may align one of surface of body element **120**, flow holes **1251-n** and **1301-n** with channel openings of a bristle assembly. By rotating collar **115**, bristle assembly **105** may be rotated and the flow holes in body element **120** allow for material to be dispensed and/or released to bristles of a hairbrush applicator. Setting or rotating to a first set of flow holes may allow for a setting, such as a light flow, and rotating to a second setting may allow for a high or regular flow of material. Positions of collar **115** may be secured and/or retained by one or more features of applicator base **110**. Applicator base **110** is shown in more detail below in FIGS. **5A-5C**. Applicator base **110** may include a rim on a top side (not shown in FIG. **1C**) including at least one guide channel allows for controlling rotation positions of collar **110**.

[0042] FIG. **2A** illustrates a hairbrush applicator with a container according to one or more embodiments. FIG. **2A** shows a front view of hairbrush applicator **200** (e.g., hairbrush applicator **100**) including bristle assembly **205**, applicator base **210**, collar **215** and container **220**. Hairbrush applicator **200** may include elements similar to those discussed in FIGS. **1A-1C**. According to embodiments, container **220** may be secured to applicator base **210** by a mechanical coupling, such as a threaded connection such that applicator base **210** may be twisted on to container **220**. Container **220** may include a chamber or reservoir to hold material, such as dry shampoo. According to embodiments, container **220** may be configured to receive a pouch or cartridge including material. Material stored and/or held in container **220** may be applied to bristles of bristle assembly **205** based on a position of collar **215** relative to applicator base **210**. According to embodiments, collar **215** may be rotated relative to applicator base **210** from at least a first position (e.g., an off position) to a second position (e.g., an on position). According to embodiments, collar **210** may be rotated from an off position, to a first on position with a light or low flow, and then rotated further to a second position with increased flow (e.g., heavy flow). When collar **215** is

rotated to an on position, hairbrush applicator **200** may be inverted and brushed through a user's hair to dispense material, such as dry shampoo to the user's hair.

[0043] According to embodiments, hairbrush applicator **200** may include one or more flow control indicators on structural elements, such as applicator base **210** and collar **215**. Flow control indicators may include signs, indicators or markings to indicate a position of collar **215** with respect to applicator base **210**, where the position indicates if one of more flow holes and channel openings are aligned to dispense material. According to embodiments, applicator base **210** includes markings **211**, **212**, and **213** and collar **215** includes indicator **216**. Alignment of indicator **216** with indicator **211** may indicate a first position, such as a closed position.

[0044] Similarly, alignment of indicator **216** with indicator **212** may indicate an intermediate position, such as a light flow setting, and alignment of indicator **216** with indicator **213** may indicate a second position, such as a regular or increased flow position.

[0045] FIG. 2B illustrates an exploded view of a hairbrush applicator with a container according to one or more embodiments. Configuration **250** illustrates includes hairbrush applicator **200**, including bristle assembly **205**, applicator base **210**, collar **215** and container **220**, with a lid **260**. According to embodiments, configurations may include lid **260** which may be removably coupled to applicator base **215**. Lid **260** may protect bristles of bristle assembly **205**. Lid **260** may also prevent leakage of material in the event hairbrush applicator **200** is not stored with collar **210** in a closed position.

[0046] As shown in FIG. 2B, an expanded view or disassembled view of hairbrush applicator **200** includes bristle assembly **205** which may be received by applicator base **210**. Collar **215** may couple with a rim of applicator base **210** and engage with the rim to securely attach to applicator base **210** and retain bristle assembly **205**. Container **220** may include threaded surface to rotatably engage and secure with the underside of applicator base **210**. FIG. 2B illustrates container **220** including a grip surface **255**. Grip surface **255** may appear on each side of the container **220** to allow for holding hairbrush applicator and as a handle. Container **220** may also be configured to store material, such as dry shampoo and personal care products. Container **220** may be a storage container including a compartment configured to store material dispensed by the hairbrush applicator and/or to store a cartridge of pre-packaged material.

[0047] Collar **215** may be configured to engage with a rim of applicator base **210** and collar **215** is configured to engage with the at least one guide channel of applicator base **210**. Collar **215** may be configured to rotate relative to applicator base **210** to provide at least a first position and a second position, the second position is configured to arrange the at least one channel opening of a base tray of a bristle assembly and at least one flow hole of a body element of applicator base **210** to dispense material to bristle assembly **205**. To couple collar **215**, collar **215** includes at least one tab protruding from an inner surface of the collar, the at least one tab configured to engage with the at least one guide channel. Collar **215** may be configured to rotate within the same plane of the at least one guide channel, and a first position of collar **215** may be configured as an off position, an intermediate position of collar **215** may be a first flow setting, and a second position of collar **215** may be a second flow setting. The second flow setting may provide increased flow of material relative to the first position.

[0048] Collar **215** may be mounted to applicator base **210** similar to a bayonet mount, wherein a rim of applicator base **210** interfaces with the interior surface or collar **215**. One or more radial pins of collar **215** may fit within a guide of applicator base **210** to keep the two parts coupled together. According to embodiments, a body element of applicator base **210** includes a plurality of flow holes having a first opening size, and a plurality of flow holes having a second opening size, wherein the second opening size is greater than the first opening size. The flow holes are each associated with a notch opening for each of the first position and the second position, and slot between the first and the second position.

[0049] One aspect of the disclosure is directed to hairbrush elements. In one embodiment, a



configuration for a hairbrush element is provided. The hairbrush element is configured for use as a component of a hairbrush. According to one embodiment the hair brush element is configured to apply a product to hair, such as a dry shampoo. Applicator base **210**, storage container **220** and collar **215** may form a handle for hairbrush applicator **200**, and bristle assembly **205** may be configured to apply at least one of dry shampoo and powder to user hair. Configurations and components discussed herein allow for application of dry shampoo to a hair brush, hair brush applicator, and storage of the hair brush and dry shampoo compartment.

[0050] As shown in FIG. **2B**, container **220** may retain a cartridge **265** which may be prepackaged. Cartridge **265** may be inserted and removed from container **220** to allow for refilling of material to hairbrush applicator **200** without requiring the user to pour powder or material into container **220**. Cartridge **265** may include a removable seal. Storage container **220** may be configured to receive and retain a removable cartridge including the material to dispense to the bristle assembly.

[0051] Another aspect of the disclosure is directed to providing alternative products to aerosol shampoo products. In one embodiment, hairbrush applicator configurations are provided as a healthier, more efficient, and environmentally friendly alternative to aerosol dry shampoo. Hairbrush applicator configurations can include one or more elements to apply powder, such as dry shampoo, directly to a bristle assembly. By brushing hair after application of the powder, the powder has minimal contact with the scalp. By using a hairbrush applicator configuration, better scalp and hair separation may be provided in order to coat the hair with the dry shampoo powder more effectively while also stripping away any residue with subsequent brushing. Hairbrush applicator configurations according to embodiments may eliminate the need to spray hair with dry shampoo.

[0052] FIGS. **3A-3B** illustrate a hairbrush applicator collar according to one or more embodiments. Collar **300** may be configured for a hairbrush applicator (e.g., hairbrush applicator **100**, hairbrush applicator **200**, etc.). According to embodiments, collar **300** includes a structure having a flattened ring with a top surface **301** and cylindrical wall **302** forming a projecting wall. Structure of collar **300** may be configured to engage with an applicator base (e.g., applicator base **110**). According to embodiments, collar **300** may be secured to the applicator base and may adjust the position of a bristle assembly when rotated. According to embodiments, collar includes a flow indicator **305** to indicate the position of collar **300**. Flow indicator **305** may be a marking or symbol on an outer surface of cylindrical wall **302**. Cylindrical wall **302** may include a flattened edge, and/or may include a curved indentation for gripping collar **300**. Interior to cylindrical wall **302** and opposite of flow indicator position **305**, collar **300** may include a pair of projections, such as projection **310**. Collar **300** may include at least one tab, such as projection **310**, protruding from an inner surface of the collar, the at least one tab configured to engage with the at least one guide channel of an applicator base. Projection **310** may include a circular outline and projection of material to engage with a guide channel of an applicator base. According to embodiments, collar **300** includes two projections, opposite from each other. Projection **310** may be located on inner surface **311** of cylindrical wall **302**. Structure of collar **300** may include engagement tab **312** extending below top surface **301**. Collar **300** may include a pair of engagement tabs, such as engagement tab **312**, located on opposite sides of collar **300** and configured to engage with a recess and/or guide element of a bristle assembly. FIG. **3B** illustrates a sectional view **350** of collar **300** along the line **3B-3B** of FIG. **3A**. As shown in FIG. **3B**, cylindrical wall **302** including projections **310** on opposite sides extending from inner surface **311**. Engagement tab **312** extends down from the top of collar **300**.

[0053] Projections **310** of collar **300** may be configured to engage with the at least one guide channel of an applicator base. When collar **300** is rotated relative to an applicator base engagement tabs, such as engagement tab **312**, the base tray of a bristle assembly is rotated to arrange the at least one channel opening of the base tray and the at least one flow hole of the body element of an applicator base to allow dispensing of material to the bristle assembly. Collar **300** may be configured to rotate within the same plane of the at least one guide channel.

[0054] FIGS. 4A-4B illustrate a hairbrush applicator bristle assembly according to one or more embodiments. FIG. 4A illustrates bristle assembly **400** (e.g., bristle assembly **105**) including a first set of bristles **405**, second set of bristles **406**, base tray **410**, and support **415**. According to embodiments, base tray **410** and support **415** provide structural elements for supporting and retaining bristles. Base tray **410** and support **415** may be joined and/or fixed together. Bristle assembly **400** may include a plurality of bristles secured to the base tray **410**, base tray **410** and support **415** each include at least one channel opening.

[0055] According to embodiments, base tray **410** includes at least one notch, such as notch **411**, configured to receive an engagement tab (e.g., engagement tab **312**) of a collar. Base tray **410** may include a pair of notches, each notch configured to receive a protrusion of a collar, such as an engagement tab of a collar, and where rotation of the collar rotates bristle assembly **400** relative to a center axis for alignment of one or more channel openings to flow holds of an applicator base. Rotation of a collar may be configured to rotate bristle assembly **400** to align the at least one channel opening of a bristle assembly with the at least one flow hole of an applicator base.

[0056] FIG. 4B illustrates an exploded view of bristle assembly **400**. Bristle assembly **400** may include a plurality of bristles including a first set of bristles **405** including an arrangement of fiber elements, and a second set of bristles **406** including a plurality of bristles each having a shaft **407** and molded tip **408**. Bristles **406** may each include a base, such as base **409** for mounting to at least one of base tray **410** and support **415**. By providing at least two types of bristles, first set of bristles **405** may be configured to distribute and evenly apply material, such as a dry shampoo and second set of bristles **406** may arrange and/or brush hair.

[0057] According to embodiments, first set of bristles **405** may be a set of fibers, set of hairs, and/or set of brush elements, wherein the base of the bristles **405** is grafted to base tray **410**. Grafting of bristles **405** to base tray **410** may allow for spacing between bristle bases. The spacing may allow for material to exit base tray, allow for other elements to be placed in proximity to the bristles, such as bristles **406**, and also improve manufacturing. Grafting of bristles **405** to base tray **410** may reduce the manufacturing time of bristle assembly **400**. Base tray **410** may be a structure made of plastic or other molded material include a plurality of openings, including channel openings **412** for material and bristle openings **413**. Bristle openings **413** may allow for pass through and/or mounting of bristles **406** through base tray **410**. One or more bristle openings **413** may be used for grafting bristles **405** to base tray. Bristles **405** may be arranged in one or more bunches or groups and bases of the groups may be fused to openings in base tray **410**. Base tray **410** may receive bristles **406** mounted from an underside of base tray **410** and through openings **413**. Support **415** may be formed of plastic, materials and general or molded of material to receive and secure a plurality of bristles, such as bristles **406** and to receive base tray **410**. Support **415** may include a plurality of openings **416** which correspond to channel opening positions of base tray **410**. Support **415** may include a plurality of projections **417** to interface with and/or retain base tray **410**. Projections **417** may extend into base tray **410**. Underside **418** of support **415** may receive material, such as dry shampoo powder, to pass through openings **416**. Underside **418** may interface and/or rest on a body support of an applicator base and allow rotation.

[0058] According to embodiments, the configuration of bristle assembly **400** allows for improvements in manufacturing. A manufacturing process may include grafting bristles to base tray **410**, inserting bristles **406** to base tray **410**, and coupling support **415** to base tray **410** to form a bristle assembly.

[0059] FIGS. 5A-5C illustrate an applicator base of a hairbrush applicator according to one or more embodiments. FIG. 5A illustrates applicator base **500** including body element **505** and rim **510**. Body elements **505** may be configured to interface with a container. Body element may also include structure, such as surface **507**, to allow for receiving a bristle assembly. According to embodiments body element **505** includes at least one flow hole to provide material to the bristle assembly. Body element **505** may include flow holes of different sizes, and a pattern of flow holes,

such that flow holes align with a position of channel openings in a bristle assembly when a collar is rotated. By way of example, body element **505** of the applicator base includes a plurality of flow holes having a first opening size, and a plurality of flow holes having a second opening size, wherein the second opening size is greater than the first opening size. FIG. 5A illustrates flow holes **508** and **509**, with flow hole **508** larger than flow hole **509**. Applicator base **500** includes rim **510** including at least one guide channel, such as guide channel **506**. According to embodiments, applicator base **500** may include a guide channel **506** on each side of body **505**. Applicator base **500** includes a notch opening for each of a first position and a second position and a slot between the first and the second position.

[0060] FIG. 5B illustrates a side view of applicator base **500**. Guide channel **506** is shown including a plurality of notches **515** and a channel **516**. FIG. 5B illustrates three notches in guide channel **506**. According to embodiments, a projection on an inner surface of a collar (e.g., projection **310**) may be positioned in a notch to arrange a bristle assembly and align flow holes with channel openings or to arrange the bristle assembly in an off position. Notches in guide channel **506** may correlate to a first position (e.g., off position), intermediate position (e.g., light flow) and second position (e.g., regular flow). Notches in guide channel may be indicated by flow indicators, such as indicator **511** for the first position, indicator **512** for an intermediate position and indicator **513** for a second position.

[0061] FIG. 5C illustrates a sectional view **520** of applicator base **500** along the line 5C-5C of FIG. 5B. As shown in FIG. 5C, applicator base **500** may include a channel guide **506** on each side of rim **510** of body element **505**. Flow opening **508** allows for an opening of to the underside of applicator base **500**.

[0062] FIGS. 6A-6B illustrate a material cartridge according to one or more embodiments. According to embodiments a hair brush applicator may be configured to receive one or more cartridges, pouches and containers of material. Using replaceable cartridges may allow for an applicator to be reused and may allow for a user to refill a container and reduce waste and/or spillage. According to embodiments, hairbrush applicators may receive a cartridge or pre-packaged refill, such as cartridge **600**. Cartridge **600** may include a sidewall **605** configured to engage with the interior of a container (e.g., container **220**). According to embodiments, sidewall **605** may include an indentation **610** to retain cartridge **600** within a container of a hairbrush applicator. According to embodiments, cartridge **600** may include seal **615** to provide a leakage protection seal. Seal **615** may include flap **616** to provide an opening in seal **615**. Seal **615** may be removable from cartridge **600**.

[0063] FIG. 6B illustrates a sectional view **620** of cartridge **600** along 6B-6B of FIG. 6A.

According to embodiments, cartridge includes compartment **625** to store material **630**, which may include dry shampoo and/or other personal care products.

[0064] While this disclosure has been particularly shown and described with references to exemplary embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the claimed embodiments.

## Claims

1. A hairbrush applicator comprising: a bristle assembly including a plurality of bristles and a base tray, wherein the plurality of bristles are secured to the base tray, and wherein the base tray includes at least one channel opening; an applicator base configured to receive the bristle assembly, the applicator base including a body element having at least one flow hole, wherein the at least one flow hole is configured to provide material to the bristle assembly, wherein the applicator base includes a rim including at least one guide channel; and a collar configured to engage with the rim, wherein the collar is configured to engage with the at least one guide channel, and wherein the

collar is configured to rotate relative to the applicator base to provide at least a first position and a second position, wherein the second position is configured to arrange the at least one channel opening of the base tray and the at least one flow hole of the body element to dispense material to the bristle assembly.

**2.** The hairbrush applicator of claim 1, wherein the plurality of bristles includes a first set of bristles including an arrangement of fiber elements, and a second set of bristles including a plurality of bristles each having a shaft and molded tip.

**3.** The hairbrush applicator of claim 1, wherein the plurality of bristles are grafted to the base tray.

**4.** The hairbrush applicator of claim 1, wherein the collar includes at least one tab protruding from an inner surface of the collar, the at least one tab configured to engage with the at least one guide channel.

**5.** The hairbrush applicator of claim 1, wherein the base tray of the bristle assembly includes at least one notch and the collar includes at least one protrusion configured to engage with the at least one notch of the base tray, and wherein the collar is configured to rotate the bristle assembly to align of the collar the at least one channel opening with the at least one flow hole.

**6.** The hairbrush applicator of claim 1, wherein the collar is configured to rotate within the same plane of the at least one guide channel, wherein the first position is configured as an off position, an intermediate position of the collar is a first flow setting, and the second position of the collar is a second flow setting, wherein the second flow setting provides increased flow of material relative to the first position.

**7.** The hairbrush applicator of claim 1, wherein the body element of the applicator base includes a plurality of flow holes having a first opening size, and a plurality of flow holes having a second opening size, wherein the second opening size is greater than the first opening size.

**8.** The hairbrush applicator of claim 1, wherein the applicator base includes a notch opening for each of the first position and the second position, and a slot between the first and the second position.

**9.** The hairbrush applicator of claim 1, wherein the applicator base is configured to engage with a storage container, the storage container including a compartment configured to store material dispensed by the applicator.

**10.** The hairbrush applicator of claim 1, wherein the applicator base and collar form a handle for the hairbrush applicator, and the bristle assembly is configured to apply at least one of dry shampoo and powder to user hair.

**11.** A hairbrush applicator comprising: a bristle assembly including a plurality of bristles and a base tray, wherein the plurality of bristles are secured to the base tray, and wherein the base tray includes at least one channel opening; a storage container including a compartment configured to store material dispensed by the hairbrush applicator; an applicator base configured to receive the bristle assembly, the applicator base including a body element having at least one flow hole, wherein the at least one flow hole is configured to provide material to the bristle assembly, wherein the applicator base includes a rim including at least one guide channel; and a collar configured to engage with the rim, wherein the collar is configured to engage with the at least one guide channel, and wherein the collar is configured to rotate relative to the applicator base to provide at least a first position and a second position, wherein the second position is configured to arrange the at least one channel opening of the base tray and the at least one flow hole of the body element to dispense material to the bristle assembly.

**12.** The hairbrush applicator of claim 11, wherein the plurality of bristles includes a first set of bristles including an arrangement of fiber elements, and a second set of bristles including a plurality of bristles each having a shaft and molded tip.

**13.** The hairbrush applicator of claim 11, wherein the plurality of bristles are grafted to the base tray.

**14.** The hairbrush applicator of claim 11, wherein the collar includes at least one tab protruding

from an inner surface of the collar, the at least one tab configured to engage with the at least one guide channel.

**15.** The hairbrush applicator of claim 11, wherein the base tray of the bristle assembly includes at least one notch and the collar includes at least one protrusion configured to engage with the at least one notch of the base tray, and wherein the collar is configured to rotate the bristle assembly to align of the collar the at least one channel opening with the at least one flow hole.

**16.** The hairbrush applicator of claim 11, wherein the collar is configured to rotate within the same plane of the at least one guide channel, wherein the first position is configured as an off position, an intermediate position of the collar is a first flow setting, and the second position of the collar is a second flow setting, wherein the second flow setting provides increased flow of material relative to the first position.

**17.** The hairbrush applicator of claim 11, wherein the body element of the applicator base includes a plurality of flow holes having a first opening size, and a plurality of flow holes having a second opening size, wherein the second opening size is greater than the first opening size.

**18.** The hairbrush applicator of claim 11, wherein the applicator base includes a notch opening for each of the first position and the second position, and a slot between the first and the second position.

**19.** The hairbrush applicator of claim 11, wherein the applicator base, storage container and collar form a handle for the hairbrush applicator, and the bristle assembly is configured to apply at least one of dry shampoo and powder to user hair.

**20.** The hairbrush applicator of claim 11, wherein the storage container is configured to receive and retain a removable cartridge including the material to dispense to the bristle assembly.

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